

DIPAK KHANDAGALE

AI/ML Engineer

Pune, India | +91 8390139357 | khandagaledipak47@gmail.com | linkedin.com/in/dipakkhandagale | github.com/Dipakk7 | dipakkhandagale.vercel.app

SUMMARY

AI/ML Engineer with hands-on experience building intelligent applications using Machine Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, and Computer Vision. Experienced in developing end-to-end AI solutions, including AI-powered assistants, resume analysis platforms, computer vision applications, and scalable backend services using Python and FastAPI. Proficient in Python, SQL, FastAPI, TensorFlow, LangChain, React, PostgreSQL, Ollama, and ChromaDB, with a strong foundation in model development, AI evaluation, REST APIs, and workflow automation. Passionate about applying AI to solve real-world problems, build reliable production-ready systems, and create impactful, user-focused solutions through intelligent automation and data-driven innovation.

TECHNICAL SKILLS

Programming: Python, SQL

AI & Machine Learning: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), Computer Vision, TensorFlow, Scikit-learn, CNN, LSTM

Frameworks & Libraries: FastAPI, React, LangChain, Pandas, NumPy, Matplotlib, Seaborn

LLM & AI Tools: Ollama, ChromaDB, Qwen 2.5

Databases: PostgreSQL, MySQL, SQLite

Deployment & APIs: REST APIs, API Integration, API Testing, JWT Authentication, OAuth2

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Power BI, Microsoft Excel, Postman

PROFESSIONAL EXPERIENCE

AI Intern | RaiTalk | Jan 2026 – May 2026

- Evaluated 50+ AI prompt variations to improve response quality, consistency, and overall model performance.
- Performed AI output evaluation, workflow validation, and functional testing across diverse real-world use cases.
- Conducted API integration testing, identified response issues, and collaborated with developers to improve application reliability.
- Validated AI-generated data and outputs to ensure accuracy, quality, and compliance with project requirements.
- Documented test results, reported defects, and supported the continuous improvement of AI models and application workflows.

PROJECTS

Scorelia – AI Career Intelligence Platform | Python, FastAPI, React, PostgreSQL, Ollama

- Built a full-stack AI career platform featuring resume analysis, ATS scoring, job matching, AI mock interviews, and personalized career roadmaps.
- Engineered a multi-agent AI system using Retrieval-Augmented Generation (RAG), LangChain, Ollama, Qwen 2.5, and ChromaDB to deliver context-aware career guidance.
- Developed secure backend services using FastAPI, SQLAlchemy, PostgreSQL, JWT Authentication, and OAuth2, and built a responsive frontend with React, Vite, and Tailwind CSS.
- Implemented automated API testing using Pytest, HTTPX, and AnyIO to improve application reliability and maintainability.

Deepfake Video Detection | Python, TensorFlow, FastAPI

- Designed and developed a deep learning-based video classification system using ResNet50 and BiLSTM to detect manipulated videos.
- Trained and evaluated the model on a dataset of 400 real and deepfake videos, achieving 85% accuracy through data preprocessing, feature extraction, and model optimization.
- Implemented face detection, frame extraction, and temporal feature learning to improve detection performance across video sequences.
- Deployed the model with FastAPI, enabling real-time deepfake detection through REST APIs.

Real-Time Face Recognition Attendance System | Python, OpenCV, YOLO, FastAPI

- Built an AI-powered attendance system using YOLO and OpenCV for real-time face detection and recognition.
- Implemented low-latency face recognition for 50+ registered users to automate attendance tracking.
- Integrated FastAPI with SQLite to manage attendance records through secure backend APIs.
- Improved recognition accuracy through image preprocessing, feature extraction, and model evaluation.

EDUCATION

Bachelor of Technology (B.Tech) – Artificial Intelligence and Data Science

MIT College of Engineering, Chhatrapati Sambhajinagar | Nov 2022 – July 2026

CERTIFICATIONS

- AI Foundations Associate – Oracle | 2025
- Generative AI Professional – Oracle | 2025
- Artificial Intelligence Fundamentals – IBM SkillsBuild | 2026
- AI/ML Virtual Internship – Kodacy & SPACE | 2026